

# **Final SOC Report: Security Monitoring & Threat Detection**

**Project:** Splunk SOC Security Monitoring Lab

**Environment:** Windows 10, Splunk Enterprise / Dashboard Studio, Sysmon v15.22, Windows Event Logs

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## **1. Executive Summary**

As part of this project, a host-based Security Operations Center (SOC) environment was designed, implemented, and tested to detect and investigate host-level security threats.

Windows Security Event Logs (``index=windows``) and Sysmon operational logs (``index=sysmon``) were integrated. Controlled attack simulations, threat hunting in SPL (Search Processing Language), and dedicated detection rules/dashboards were built for two primary incident scenarios.

## **2. Architecture & Environment Setup**

### **1. Windows Security Log Ingestion:**

Ingestion of failed (``EventCode=4625``) and successful (``EventCode=4624``) authentication events.

Field mapping targeting native Windows log field names (``Account_Name``, ``Source_Network_Address``).

### **1. Sysmon Installation & Configuration:**

Installed Sysmon service and applied the SwiftOnSecurity rule set (``sysmonconfig-export.xml``).

Enabled process creation (``EventCode=1``) and outbound network connection (``EventCode=3``) tracking. \* Routed events to the dedicated ``sysmon`` index via ``inputs.conf`` configuration.

### 3. Incident Investigation & Detection

#### ● Incident 001 — Brute Force / Password Spraying Attack

**Scenario:** The attacker executed a password guessing/spraying attack against user accounts ( `admin`, `bob`, `karol`, `maciek` ), culminating in a successful account breach.

**Source IP:** `127.0.0.1` (localhost loopback / compromised endpoint).

**Observed Events:** 19 failed logon attempts and 71 successful logons.

#### ● Incident 002 — Suspicious PowerShell Execution & Network Connection

**Scenario:** Execution of PowerShell with security-bypass switches ( `-ExecutionPolicy Bypass`, `-NoProfile` ) and an outbound web request ( `Invoke-WebRequest` ).

**Correlation Analysis:** Correlated the spawned process ( `EventCode=1` ) with its corresponding outbound network traffic ( `EventCode=3` ) using the shared `ProcessId`.

**Noise Reduction:** Filtered out legitimate internal Splunk processes ( `splunk-powershell.exe` ).

### 5. Summary & Key Takeaways

**1. Telemetry Verification:** Sysmon provides process execution and network connectivity context that standard Windows Event Logs lack.

**2. Field Extraction Precision:** Event fields vary across sources ( `Account\_Name` in Windows Security vs `User` in Sysmon). Proper grouping ( `stats by` ) and handling null values ( `coalesce` ) prevents false-negative query outputs.

**3. SOC Deployment Ready:** Practical detection queries and panel layouts were established for deployment inside SOC Security Monitoring Dashboards.